

Clinical, immunological, and microbiological evaluation of experimental peri-implant mucositis and gingivitis in subjects with Grade C, stage III/IV periodontitis background.

Running title: Gingivitis/peri-implant mucositis in Grade C, stage III/IV periodontitis background.

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Author contributions

T.P.D. and E.A.S conceived/designed the study, contributing to data collection, analysis/interpretation, drafting, editing, and revising the manuscript. I.L.F.G. obtained all the clinical parameters and performed sampling. M.F.M, R.C.V.C and P.S.K contributed to data analysis/interpretation, preparation of figures and editing/revising the manuscript. M.Z.C and K.G.S.R contributed to data interpretation/discussion and editing/revising the manuscript. All authors approved the final version of the manuscript.

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ABSTRACT

Aim: To compare individuals with a periodontitis background (Grade C, stage III/IV - former generalized aggressive periodontitis) (H-GAP) to periodontally healthy subjects (H-Health) in terms of molecular changes (immunological/microbiological) accompanying experimental peri-implant mucositis and gingivitis.

Material and Methods: H-GAP and controls (H-Health) subjects were recruited, and experimental mucositis/gingivitis induced around a single screw-retained implant and one contralateral tooth. Participants refrained from oral hygiene for 21 days in the selected areas, followed by professional prophylaxis and hygiene instructions for 21 days. Clinical parameters, immunological markers (multiplex analysis) and microbial data (16S rRNA gene sequencing) were collected at baseline, during induction (7,14, and 21 days) and following remission (42 days).

Results: Clinically, no significant differences were observed between groups (n=10/each group) (H-GAP versus H-Health) ($p>0.05$, Mann-Whitney test) and type of site (tooth versus implant) ($p>0.05$, Wilcoxon test) in the time of onset and resolution of, or severity of gingival/mucosal inflammation. H-GAP displayed a lower concentrations of the cytokines IL-1B, IL-4, IL-17, TNF-A, INF-G around implants than H-Health at baseline and during induction of mucositis ($p<0.05$, Mann-Whitney test). In both groups, implants demonstrated significantly higher inflammatory background at baseline and all subsequent visits when compared to teeth ($p<0.05$, Wilcoxon test). Alpha and beta-diversity metric demonstrated a significant shift in the microbiome composition and abundances of core species during induction and resolution of peri-implant mucositis and gingivitis ($p<0.05$, REML of Shannon and Bray-Curtis indices respectively). Differences were not significant for these parameters between H-Health and H-GAP groups when the periodontal and peri-implant microbiomes were compared separately, however, at each time point the peri-implant microbiome differed significantly from the periodontal microbiome.

Conclusion: Within the limitations of this pilot study (e.g., low power), it can be concluded that different microbial shifts contribute to onset and progression of inflammatory responses around teeth and implants, and that history of periodontal disease experience plays an additional role in modulating the immune response of peri-implant and periodontal tissues to biofilm accumulation.

Keywords: Periodontal disease; gingivitis; dental implants; Inflammation; crevicular fluid;

microbiome.

Clinical Relevance

Scientific rationale for the study: The microbiological and immunological features associated with experimental peri-implant mucositis/gingivitis in patients with Grade C stage III and IV periodontitis background (former generalized aggressive periodontitis) is not fully understood.

Principal Findings: Implants present with a higher inflammatory background than teeth even in states of health, and the pro-inflammatory responses of the peri-implant soft tissues to progressive biofilm accumulation are of higher magnitude than gingival tissues. However, a history of periodontitis predisposes to a dampened inflammatory response to biofilm accumulation around implants. Taken together, the findings suggest that the peri-implant tissue in individuals with a periodontitis background responds differently to microbial plaque than periodontally healthy individuals.

Practical Implications: Although clinical responses to plaque accumulation do not capture the magnitude of molecular changes, responses of implants differ from teeth in individuals with a history of periodontitis, emphasizing the need for further studies on long term care and maintenance of implants, especially in patients with periodontitis.

INTRODUCTION

Tooth loss is a serious health problem that affects quality of life (GBD, 2019; WHO Global Oral Health Status Report, 2022). Thus, the introduction of dental implants to replace missing teeth became one of the great achievements in dentistry. Despite the high success and survival rates of implant therapy, complications associated with infection and inflammation do occur.

According to the literature, the factors associated with peri-implant diseases are poor oral hygiene, cigarette smoking, diabetes, alcohol consumption and genetic traits (Heitz-Mayfield LJA, 2008; Sgolastra F et al, 2013; Pimentel SP et al 2018), with history/background of periodontitis considered the strongest predictor of peri-implantitis (Anderson N et al, 2020). Patients with a history of aggressive periodontitis (Armitage GC, 1999), are reported to have higher incidence of bone loss and peri-implantitis and lower rates of implant success and survival (Mengel & Flores-de-Jacoby, 2005; Mengel et al, 2007; De Boever et al, 2009; Swierkot et al, 2012; Theodoridis C et al, 2017). Therefore, although implant therapy is a reliable and safe technique, rehabilitation of patients with a history of aggressive periodontitis remains a challenge for researchers and clinicians and there is an urgent need to better understand the factors that contribute to peri-implant disease in these individuals.

Experimental gingivitis has been long established as a model to examine the dynamics of biofilm-induced inflammation. This model has also been extended and validated to study the development and resolution of inflammation in the peri-implant sulcus of periodontally healthy subjects (Pontoriero R et al, 1994; Zitzmann NU et al, 2001; Schwarz F et al, 2014; Schincaglia GP et al, 2017; Ribeiro FV et al, 2017; Clever K et al, 2019; Meyer S et al, 2019). Additionally, it was previously reported that in experimental gingivitis trials, patients with a history of periodontitis demonstrated a higher rate of plaque accumulation and earlier development of redness and bleeding on probing when compared with periodontally healthy individuals (Van der Velden U et al, 1985; Abbas F et al, 1986; Johnson TC et al, 1996). Furthermore, in periodontally healthy patients, the progression events of peri-implant mucositis were similar to those of experimental gingivitis but led to a different clinical, immunological and microbiological response (Nagasawa MA et al, 2022). However, within the authors' knowledge, there is no study comparing tooth and dental implant in patients with different periodontal backgrounds using a controlled and reversible method that enable examining the clinical, microbiological, and host-response shifts.

Hence, the primary aim of the present investigation was to compare individuals with a periodontitis background (Grade C, stage III/IV - former generalized aggressive periodontitis) (H-GAP) to periodontally healthy subjects (H-Health) in terms of molecular changes (immunological/microbiological) accompanying experimental peri-implant mucositis and gingivitis. While the secondary aim was the comparison of dental implant versus tooth (split-mouth) in terms of microbiological, immunological, and clinical events associated with experimental mucositis/gingivitis.

MATERIALS & METHODS

This was a single-center, experimental gingivitis/mucositis pilot study registered in the U.S. National Institutes of Health Clinical Trials Registry (NCT03713567), with parallel and split-mouth comparisons. The study was approved by the Ethical Committee of the Piracicaba Dental School – State University of Campinas, Piracicaba, Brazil (CAAE #70803817.1.0000.5418) and conducted in accordance with the Helsinki Declaration of 1975, and the observational studies in epidemiology (STROBE) guidelines. All participants were recruited between November 2017 and February 2018 from patients previously treated and under periodontal/peri-implant maintenance at the Piracicaba Dental School – State University of Campinas. The participants were informed about the procedures and signed the informed consent form.

Participant enrollment and Clinical Protocol

The study design compared the changes occurring around implants and teeth during the experimental mucositis and gingivitis trial in grade C stage III and IV periodontitis background patients and subjects with no history of periodontitis. The study inclusion criteria were age ≥ 18 years, and good general health (American Society of Anesthesiologists (ASA) I or II). Individuals who were previously diagnosed for generalized aggressive periodontitis (Armitage GC, 1999), and who presented with ≥ 20 teeth at the time of recruitment; and under semiannual supportive periodontal/peri-implant care were included in the H-GAP group. According to the new classification of Periodontal Diseases, the patient's clinical and radiographic characteristics of the H-GAP group is now assigned as Periodontitis Grade C stage III or IV. Periodontally healthy subjects, with no history of periodontitis, were included in the H-Health group. Exclusion criteria were current or immunosuppressive therapy; use of antibiotics, anti-inflammatory drugs, or any medication known to affect periodontal status within three months of enrollment; patients with major systemic diseases such as diabetes mellitus and cancer; smokers; pregnant and lactating women. For the implant selection, 1 single implant (Straumann® system) rehabilitated with a screwed metal-ceramic crown in function for at least 6 months was included. For the teeth sites, it was included 1 tooth located at the same arch in the corresponding position of the implant. Both (tooth and implant) included sites presented no probing depth $>4\text{mm}$; no clinical attachment level $>5\text{mm}$, no interproximal alveolar bone loss, and Plaque Index $<20\%$ (Silness e Loe, 1964). Sites with peri-implantitis (Lang NP et al, 2011) and/or periodontitis (Armitage GC, 1999) in possible test areas were excluded. Due to the innovative objective of the present pilot study, with strict inclusion and exclusion criteria, the sample size was estimated based on previous investigations with similar parameters (Salvi GE et al, 2011; Meyer S et al, 2016; Schincaglia GP et al, 2017). However, post-hoc power analysis using paired t-test at H-GAP and H-Health with the delta of the modified gingival index revealed a statistical power of 33.8% (Supplementary material).

Figure 1 shows the flow diagram for the three phases of the study. During the preparatory phase, informed consent and medical/dental history was obtained, following which, supportive periodontal/peri-implant therapy (SPT) including polishing and oral hygiene instructions was provided to the participants. Personalized acrylic stent was fabricated to cover approximately 2mm of the soft tissues with no contact over the soft tissues in the test areas. In order to begin the second phase (day 0) and receive the customized stents with instructions of use, all participants had to present an angulated bleeding score (ABS) (Van der Weijden GA et al, 1994) equal to zero. The participants wore the removable stents to cover the test areas (one

tooth and one dental implant) during daily brushing procedures. The periodontal/peri-implant parameters: modified gingival index – mGI (Trombelli L et al, 2004), angulated bleeding score – ABS (Van der Weijden GA et al, 1994), plaque index – PI (Silness J and Loe H, 1964), gingival crevicular fluid – GCF - volume and peri-implant crevicular fluid – PICF - volume were recorded; crevicular fluid and plaque sampling were also collected weekly at days 0, 7, 14 and 21 by a single calibrated examiner (ILFG), examiner's Intraclass Correlation Coefficient equals to 0.945. At the beginning of the resolution phase, the subjects received a professional prophylaxis and instructions to clean the tooth and implant properly. The subjects were further weekly monitored for safety and a professional prophylaxis was performed. At day 42, the periodontal exams, GCF/PICF and plaque sampling were repeated followed by professional prophylaxis.

Sample collection

The procedure for sample (plaque, GCF and PICF) collection was kept consistent throughout the trial (days 0, 7, 14, 21, and 42) and the same tooth and dental implant included in the study. To get the most representative scenario of the host-bacterial interaction without disturbing the biofilm, the sample collection was performed on an undisturbed (non-repeated) surface (buccal, mesio-buccal, disto-buccal, lingual, mesio-lingual, disto-lingual), at each time point of the trial. All test sites were isolated with cotton rolls and dried. Samples of GCF and PICF were collected with paper filter strips (Periopaper®, Oraflow, Smithtown, NY®). The Periopaper® was left in place for 30 seconds. The fluid volume was determined (Periotron 8000, Oraflow, Plain view, NY, USA®) and paper filters were stored in a microtube (coded for each participant) at -20°C, until analyzed. The subgingival biofilm was collected using sterile paper cones (Tanariman Industrial LTDA, Manacapuru, AM, Brasil). The paper cones were inserted to the bottom of the sulcus, left in place for 30 seconds, then removed and stored in a micro-tube at -20°C until analyzed.

Immunological analysis

The cytokine levels of Interleukin (IL)-10, IL-1 β , IL-4, tumor necrosis factor (TNF)- α , interferon (INF)- γ , IL-6, IL-17 and Osteocalcin (OC), Osteopontin (OPN), Osteoprotegerin (OPG) and RANK-L in the GCF and PICF were determined via a multiplexed fluorescent bead-bead immunoassay (HCYTOMAG-60K-07, HBNMAG-51K-03 and HRNKLMAG-51K-01 kits, Millipore Corporation, Billerica, MA, USA). Assays were carried out according to the manufacture's recommendations using the Magpix™ instrument (MiraiBio, Alameda, CA). All

cytokine values were expressed as a concentration (pg/ml).

Microbiome analysis

Bacterial DNA was isolated using a Qiagen MiniAmp Kit (Valencia, CA) according to the manufacturer's instructions and stored at -20°C. The hypervariable V4 (515/806) region of the 16S rRNA gene was amplified and sequenced on the Illumina Miseq platform to produce 250bp paired-end sequences.

Bioinformatic analysis was performed using the QIIME 2 2019.7 (Bolyen E et al, 2019). Taxonomy assignment of amplicon sequence variants (ASVs) was performed using the expanded Human Oral Microbiome (eHOMD, V15.2) Database (Escapa et al, 2018). Alpha-diversity was analyzed using the Shannon metric (Spellerberg IF, 2003) and beta-diversity analysis was performed using the Compositional Tensor Factorization (Martino C et al, 2021) and visualized with PCoA (Principal Coordinate Analysis). Furthermore, Bray-Curtis, Jaccard, Unifrac Weighted and Unifrac Unweighted metrics were used to estimate beta-diversity of each group/site across time, separately. Linear Discriminant Analysis (LDA) was used for dimensionality reduction. LDA plots were generated with PhyloToAST (PCoA.py). The core microbiome was considered as species presented in at least 50% of the samples from a group. The cytokines concentration and the relative abundance of the most important selected species were analyzed using linear discriminant analysis (LDA). For a more detailed methodological description of bacterial analysis, please see Supplementary material.

Data Analysis

GraphPad Prism software 9.3.0 (1992-2021 GraphPad Software, LLC[©]) was used to perform all demographic, clinical and immunological data analysis. The data distribution was analyzed using the Shapiro-Wilk test. Comparison between groups (H-GAP versus H-Health) at a single time point was evaluated via unpaired-samples, Mann-Whitney test. Differences between tooth and implant sites at a single time point were evaluated via paired-sample Wilcoxon tests. The responses of tooth and implant sites during experimental phase (0, 7, 14, 21 and 42) were evaluated with Friedman test, and to evaluate 2-time points, the Dunn test was performed. All tests considered alpha=5%. Since microbiome data demonstrated variances over time, all microbial community parameters (e.g., alpha- and beta-diversity) were statistically tested with the restricted maximum likelihood method (REML) to reduce biases in variance and co-variance.

RESULTS

Twenty subjects (10 per group) gave informed consent and were enrolled in the study. During the clinical protocol, no complications or adverse events were reported. Among the 20 implants included, none presented mucosal recession. Among the 20 teeth included, small gingival recessions were observed ($\leq 3\text{mm}$ / RT1 A-) (Cairo et al 2011). The clinical parameters at baseline are shown in Table S1. Among the 20 individuals who started the study, none were excluded or dropped out. The H-GAP (n=10) and H-Health (n=10) groups were not different demographically ($p>0.05$, 2-sample t-test, Table S1- Supplementary Material).

Baseline characteristics of teeth and implants in H-GAP versus H-Health individuals: At baseline, implants and teeth did not differ clinically ($p>0.05$, Mann-Whitney test, Table 1) or microbiologically ($p>0.05$, REML test, Figure 3) between H-GAP and H-Health groups. However, implants from H-GAP group displayed lower levels of IFN- γ , IL-10, IL-17, IL-1 β , IL-4 and TNF- α compared to H-Health ($p<0.05$, Mann-Whitney test, Figure 2), while teeth from both groups did not demonstrate significant differences in the measured cytokines. Moreover, implants showed higher concentrations of cytokines when compared to teeth in both groups ($p<0.05$, 2-sample Wilcoxon-test, Figure 2).

Induction and resolution of experimental peri-implant mucositis - H-GAP versus H-Health: Despite the similar trends in immunological response during the induction and resolution of experimental mucositis (Figure 2), significant differences were observed between H-GAP and H-Health groups. During the induction phase IFN- γ , IL-10, IL-17, IL-4, TNF- α , and IL-1 β were significantly lower at implants of H-GAP group compared to H-Health ($p<0.05$, Mann-Whitney test, Figure 2). Additionally, during the resolution phase of experimental mucositis, IL-17, and IL-1 β was also lower at implants of subjects from H-GAP group ($p<0.05$, Mann-Whitney test, Figure 2). These differences in immune markers were not accompanied by differences in clinical parameters, during the induction and resolution of experimental mucositis, no significant difference was observed between H-GAP and H-Health at any of the indices analyzed (PI, mGI, ABS, CF) ($p>0.05$, Mann-Whitney test, Table 1). Similarly, analysis based on α -diversity (Shannon metric), no difference between H-GAP and H-Health was observed (Figure 3A, REML test, $p>0.05$). Furthermore, in the analysis based on β -diversity, statistically significant difference among the communities of each time-point was observed (Figure 3, B, CTF, D a-d, Bray-Curtis; e-h, Jaccard; i-l, Unifrac Unweighted; m-

p, Unifrac Weighted), however no significant difference between the groups H-GAP and H-Health was observed ($p>0.05$, REML test, Figure 3B,C).

Induction and Resolution of experimental inflammation - Tooth versus Implant: In both H-GAP and H-Healthy groups, teeth and implants evidenced significantly different microbial shifts ($p<0.05$, REML test, Figure 3) and immunological responses ($p<0.05$, Friedman test, Figure 2) to these shifts during induction and resolution of inflammation.

During the induction phase, the IFN- γ , IL-17, IL-4 and TNF- α concentrations around implants were significantly higher compared to teeth in both groups ($p>0.05$, 2-sample Wilcoxon-test, Figure 2), while at the H-Health IL-10 and IL-1 β were also significantly higher at implants compared to teeth ($p>0.05$, 2-sample Wilcoxon-test, Figure 2)

Microbiologically, figure 3 demonstrates the changes in the microbial diversity for tooth and implant, in both groups. Fig. 3A shows a significant variation across time of the alpha (α)-diversity (Shannon index) in the teeth sites of both groups. No significant difference was observed over time in implants. Regarding the beta-diversity analysis, statistically significant shifts ($p < 0.05$ REML test, Figures 3 B-D) across time was observed in both teeth and implants. The PCoA (Figure 3B) shows the microbiological dissimilarities during biofilm accumulation and resolution phase in teeth and implants of both groups. The changes across time in teeth were more evident than in implants, regardless of the group (Figure 3B and D, a-p). Additionally, figure 3C presented the longitudinal β -diversity variation (first axis – PC1), showing that both implant and tooth changed significantly across time. However, during the induction phase, implants demonstrated a significantly lower diversity ($p < 0.05$, REML, Figure 3B) when compared to teeth (CTF metric). Following clinical resolution of inflammation (Day 42), the peri-implant microbiome returned to a similar profile as the baseline, however, the periodontal microbiome reset to a significantly lower diversity (CTF metric) than baseline ($p < 0.05$, REML, Figure 3C).

Inflammatory and microbiological characteristics of Tooth and Implant Sites: Figure 4 shows the summary association between the bacteria (species-level) and cytokines level of tooth and implant sites using Linear Discriminant Analysis (LDA) at clinical health (days 0 and 42), and clinical inflammation (days 7, 14, and 21) irrespective of the history of periodontitis. The analysis described a different inflammatory and microbial pattern between tooth and implants that were mostly discriminated by the inflammatory markers, where the tooth was related to IL-17, IL-10, and TNF- α , and implants were related to IFN- γ , IL-4. Additionally, the implant site presented an overlapping of clinical health and mucositis, while a more evident distance between clinical health and gingivitis was observed in tooth sites.

4 DISCUSSION

The present pilot study explored differences between individuals with a background of periodontitis and individuals with periodontal health that were submitted to experimental plaque accumulation around one dental implant and tooth. Therefore, in addition to group comparisons, the design of the study allowed comparisons between different type of sites (implant versus tooth) in the same individuals (split-mouth). Within the limitations of this experimental conditions (e.g., low power), it was observed that the immune response of peri-implant and periodontal tissues may be driven by site and periodontal background, while the

microbiological response was affected by the type of site. However, no significant clinical differences between tooth and implant and between groups were demonstrated. In this respect, it can be noted that a recent systematic review of similar studies presents a number of included participants up to 25 (Nagasawa MA et al, 2022). In spite of that, care should be exercise when interpreting the results of this pilot study, since the post-hoc power analysis revealed low power. The approach of matching different periodontal diagnosis and different sites under strict inclusion criteria favored the comparisons but also was expected to limit the number of eligible participants. Therefore, lack of significant differences and disagreements with previous studies should be viewed with caution and be addressed in larger studies. Nevertheless, concerning the comparison between type of sites, the absence of clinical differences is consistent with the existing literature (Pontoriero R et al, 1994; Zitzmann NU et al, 2001; Schincaglia GP et al, 2017) that also demonstrated no significant differences in the clinical parameters between tooth and implant, after biofilm accumulation. On the other hand, previous studies reported that patients with a periodontitis diagnosis compared to non-periodontal subjects, demonstrated an increased rate of plaque accumulation associated with early development of redness, bleeding on probing, and a higher gingival crevicular fluid volume during the experimental gingivitis. (Van der Velden U et al, 1985; Abbas F et al, 1986; Johnson TC et al 1996; Trombelli L et al, 2006).

Although inflammation-associated clinical signs occurred, there were no significant increases at cytokines levels during the biofilm accumulation. Albeit this result could be apparently contradictory, Offenbacher S et al (2010) reported a decrease in several cytokines during the experimental gingivitis. The authors suggested that this phenomenon may represent a degradation of cytokines within the sulcus or a difference or shift in chemokine compartmentalization in tissue versus crevicular fluid. Additionally, an in-vitro study with *P. gingivalis* demonstrated that bacterium may be able to release an endotoxin that may suppress chemokine synthesis compared to the stimulatory activity of enteric LPS species (Darveau RP et al, 1998). It should be noted that the absence of statistically significant differences or the discrepancies in the inflammatory markers results may be due to variability across individuals in the concentrations of inflammatory markers (high, low, and slow responders) (Bamashmous S et al, 2021), and the number of participants (Ghallab NA et al, 2018), therefore should be cautiously considered.

In the present study, significant differences between day 42 and the plaque accumulation phase were observed for some cytokines. Indeed, the participants went through a very rigid plaque control after induction phase, including 2 sessions of prophylaxis within 21 days,

resulting in complete control of any clinical inflammatory signs. It should be recognized that this professional plaque control may have influenced the clinical, immunological, and microbiological response and should be considered to compare the present results with other studies. Regarding some cytokines concentration increases at 42 days, it is interesting to note that a study exploring the microbiota and inflammatory characteristics during the resolution of naturally occurring gingivitis did not observed changes in the inflammatory mediators after the hygiene phase and professional treatment (Stone et al, 2018). Thus, the results showed that the time-course of cytokine release – and the reduction of it – during inflammation establishment and resolution, still need further clarification. It is possible that there is a time-dependency between the stimulus and changes in GCF/PICF local levels, and the role of each cytokine acting as a stimulus for other inflammation player, during disease occurrence. Moreover, the biological relevancy of each cytokine local concentration also remains poorly understood and future studies are needed to be able to better address these aspects.

Immunological analysis showed a difference between the responses of subjected-matched implant and tooth sites and between H-Health and H-GAP subjects. Implants of both groups showed higher concentration of inflammatory and anti-inflammatory cytokines compared to teeth. Recently, Serichetapongse P et al (2022) proposed that the presence of the implant and the abutment materials may influence the level of inflammatory cytokines in PICF. Therefore, it can be hypothesized that the materials and anatomical differences between implants and teeth may have an influence on the concentration of cytokines. Consequently, these results should be further explored in a bigger population with different periodontal backgrounds, aiming to understand the individual variability observed in the literature, and the impact of the titanium surface, as well as the anatomy and prosthetic materials in the crevicular fluid content. In addition, a significantly lower concentration of most of the cytokines analyzed was observed at implants of the H-GAP group when compared to H-Health, which may suggest a depressed and maybe less complete inflammatory response of the H-GAP group, in response of a similar clinical stimulus and microbiological community. Hence, these results should be further explored to verify if this apparently depressed immunological response should be one of the reasons for the susceptibility of periodontal patients to peri-implants diseases. Similarly, Elabdeen HRZ et al (2017), observed significant lower levels of cytokines at GCF and plasma of Generalized Aggressive Periodontitis individuals compared to healthy controls. The authors suggested that the great individual variability of cytokines observed in plasma may be due to differences in the ability to establish an adequate inflammatory response. Therefore, a proper balance between destructive and protective roles of cytokines in periodontitis may protect the

periodontal tissues (Garlet GP et al, 2010). The biological meaning of the differences in cytokine levels observed in the present study needs further evaluation in additional larger studies, addressing individual variability and possible impact of factors like titanium surface, anatomical differences, and patient periodontal background in this complex immunological process.

It was interesting to note that, despite the lack of significant increase in the CF content during mucositis/gingivitis induction, a shift in the bacterial community and an increase in the microbiome core of species at day 21, at all sites and groups compared with days 0 and 42 was observed. This result suggests maturation of the microbial community during inflammatory induction and the increase in complexity, with the emergence of later colonizers. Additionally, it was possible to note the elevated percentage of species shared between sites, despite of the groups which corroborates with the alpha and beta diversity analysis, suggesting that the periodontal history of the subject did not affect the microbiological response under the conditions of the present study.

The difference between sites in the β -diversity (CTF metric), corroborates with Heuer W et al (2012), who identify different bacterial compositions at inflamed implant and tooth sites in the same individual, with a high diversity in tooth. Similarly, Schincaglia GP et al (2017) observed a more evident increased microbiome diversity across time in tooth than implants. Thus, the peri-implant environment seems to respond differently, being more resistant to change than the tooth during inflammation establishment. Therefore, when comparing the community changes around dental implants and teeth, it could be suggested a resilience in the microenvironment in the tooth compared to higher resistance in the peri-implant tissues. Although the reason for that phenomenon is still not fully known, it can be speculated that the implant titanium surface, as well as the anatomical, physiological and inflammatory differences of peri-implant environment compared to the periodontium, can significantly interfere in this process (Heuer W et al, 2012). These dissimilarities driven by the local characteristics of each site, could impact on their behavior during inflammatory induction and its resolution.

Additionally, integrative microbiome-inflammatory data analysis highlighted the difference between teeth and implants sites during the clinical healthy and inflamed phases. Implant presented a more stable response during experimental peri-implant mucositis even when microbiome and cytokines were evaluated together. In opposite, teeth sites presented an evident response to the clinical phase. Interestingly the most descriptive variables to discriminate the sites and clinical phase were the cytokines, being IL-10, IL-17 and TNF- α , the most discriminant of the tooth and, IFN- γ and IL-4 for the implant. These results emphasize the

importance of inflammation in modulating the subgingival/peri-implant environment, at the same time they demonstrate how only species/genera were not good descriptors of the subgingival/peri-implant site. Interestingly, the previous studies showing a higher prevalence of inflammatory disorders around implants installed in periodontal patients led us to hypothesize an elevated concentration of cytokines in the PICF environment during mucositis in H-GAP compared to individuals with no history of periodontitis. However, under the conditions of the present study, the analysis showed what seems to be a depressed inflammatory response in the H-GAP group that could not be attributed to microbiological composition differences. In addition, it also showed a consistent effector aspect of implant on the local environment over the immunological and microbiological shifts.

It is important to highlight that previous studies and the present one considered the 1999 Classification of Periodontal Diseases (Armitage, 1999) to select the subjects, which was based on the evidence that subjects with a diagnosis of Generalized Aggressive Periodontitis show a specific family-related immunological and microbiological profile. However, in face of the newest classification of periodontal diseases and considering clinical and radiographic characteristics of the individuals, the authors adapted the term to the new one in order to favor presentation and current understanding of the results observed.

In spite of unprecedented associations in the comparisons of periodontal backgrounds and between tooth and dental implants, suggesting that the peri-implant tissue in individuals with a periodontitis background responds differently to microbial plaque than periodontally healthy individuals, it is worth highlighting the exploratory nature and limitations of this study. It can be assumed that the differences between teeth and implants (e.g., surface roughness, free energy, chemical composition and implant abutments) are limitations, even with the effort of standardization of the type of implant and clinical parameters around the teeth and implants. The heterogeneity in the subject-specific and site-specific (e.g., presence of small gingival recession) characteristics associated with a well-known variability across individuals in the microbiome profiles and concentrations of inflammatory markers should be kept in mind. In addition, the strict inclusion criteria and sample size (study power) might have influenced the results.. Hence a larger study would be necessary to confirm some of the trends observed in the between-group analyses and the unique associations uncovered in the split-mouth (tooth versus implant) comparisons.

5 CONCLUSION

Within the limitations this pilot study (e.g., low power), it can be concluded that the

immune response of peri-implant and periodontal tissues to the biofilm accumulation were driven by site and periodontal history, while the microbiological changes were affected mainly by the type of site (implants versus teeth).

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Tables

Table 1. Clinical parameters before (day 0) during (days, 7, 14, 21) and after (day 42) the induced biofilm accumulation, expressed as median and inter-quartile range.

		H-GAP			H-Health			H-Health vs H-GAP	
		Tooth	Implant	<i>p</i> -value Tooth vs Implant	Tooth	Implant	<i>p</i> -value Tooth vs Implant	Tooth <i>p</i> -value	Implant <i>p</i> -value
PI	Day 0	0 (0.0-1.0)	0 (0.0-1.0)	0.250	0 (0.0-1.0)	0.5 (0.0- 1.0)	0.687	0.521	>0.99
	Day 7	1 ^δ (1.0-2.0)	1 ^δ (1.0-1.0)	0.250	2.0 ^{*δ} (1.0-2.0)	1.5 ^δ (1.0-2.0)	0.500	0.178	0.140
	Day 14	2 ^{*δ} (1.37-2.0)	1 ^δ (1.0-2.0)	0.109	2.0 ^{*δ} (2.0-2.0)	1.5 ^δ (1.0-2.0)	0.125	0.791	0.649
	Day 21	2 ^{*δ} (1.87-2.0)	1.75 ^{*δ} (1.0-2.0)	0.125	2.00 ^{*δ} (2.0-2.0)	2.00 ^{*δ} (1.75- 2.0)	0.625	0.945	0.499
	Day 42	0 (0.0-0.0)	0 (0.0 - 0.0)	>0.99	0 (0.0- 0.0)	0 (0.0-0.0)	>0.99	>0.99	>0.99
	<i>p</i> -value	< 0.0001	< 0.0001		< 0.0001	< 0.0001			
mGI	Day 0	1 (0.37-1.0)	0.75 (0.0-1.0)	0.437	0 (0.0-1.0)	0 (0.0-1.0)	>0.99	0.169	0.432
	Day 7	1 ^δ (1.0-2.0)	1 ^{*δ} (1.0-1.0)	0.250	1 (1.0-1.25)	1 ^{*δ} (1.0-2.0)	>0.99	0.628	0.582
	Day 14	1 ^δ (1.0-2.0)	1 ^{*δ} (1.0-1.12)	0.500	2 ^{*δ} (1.0-2.0)	1 ^δ (1.0-2.0)	0.250	0.293	0.582
	Day 21	2.0 ^δ (1.37-2.0)	1.75 ^{*δ} (1.0-2.0)	0.250	2 ^{*δ} (1.0-2.0)	2 ^{*δ} (1.0-2.0)	>0.99	0.742	0.649
	Day 42	0 (0.0-0.0)	0 (0.0-0.50)	0.500	0 (0.0-0.25)	0 (0.0-0.25)	>0.99	0.473	>0.99
	<i>p</i> -value	< 0.0001	< 0.0001		< 0.0001	< 0.0001			
ABS	Day 0	0 (0.0-0.0)	0 (0.0-0.0)	>0.99	0 (0.0-0.0)	0 (0.0-0.0)	>0.99	>0.99	>0.99
	Day 7	0 (0.0-1.0)	0 (0.0-0.25)	0.500	0 (0.0-1.0)	0 (0.0-1.0)	>0.99	>0.99	0.628
	Day 14	0 (0.0-1.0)	0 (0.0-0.50)	0.375	1 (0.0-1.0)	0 (0.0-1.0)	0.500	0.656	0.442
	Day 21	1 (0.37-1.0)	0.75 (0.0- 1.0)	0.250	1 (0.0-1.0)	1 (0.0-1.0)	>0.99	0.828	0.6499
	Day 42	0 (0.0-0.0)	0 (0.0- 0.0)	-	0 (0.0-0.0)	0 (0.0-0.0)	-	>0.99	>0.99
	<i>p</i> -value	0.0014	0.0025		0.0037	0.0026			
CF (µl)		GCF	PICF	<i>p</i> -value GCF vs PICF	GCF	PICF	<i>p</i> -value GCF vs PICF	GCF	PICF
	Day 0	0.20 ^δ (0.11–0.33)	0.31 (0.19–0.52)	0.310	0.13 (0.09– 0.16)	0.12 (0.05 – 0.31)	0.503	0.146	0.092
	Day 7	0.18 ^δ (0.11–0.43)	0.29 (0.08– 1.37)	0.492	0.17 (0.13–0.22)	0.26 (0.17– 1.66)	0.250	0.985	0.668
	Day 14	0.16 (0.10–0.25)	0.21 (0.12–0.96)	0.314	0.35 ^δ (0.21– 0.44)	0.49 (0.32– 1.40)	0.322	0.060	0.210
	Day 21	0.30 ^δ (0.12– 1.66)	1.26 ^δ (0.27– 1.87)	0.312	0.35 ^δ (0.13–0.82)	0.44 (0.21 – 1.67)	0.232	0.752	0.301
	Day 42	0.07 (0.05–0.09)	0.10 (0.05 – 0.21)	0.117	0.07 (0.06 – 0.12)	0.10 (0.06–1.17)	0.539	0.694	0.927
	<i>p</i> -value	0.0090	0.0079		0.0005	0.0183			

Note: Plaque Index (PI), Modified Gingival Index (mGI), Angulated Bleeding Score (ABS), Gingival Crevicular Fluid (GCF) volume and Peri-Implant Crevicular Fluid volume (PICF). Mann-Whitney test

for comparison between groups (H-GAP versus H-Health) at a single time. Differences between tooth and implant sites at a single time point were evaluated via paired-sample Wilcoxon tests. The longitudinal analysis (0, 7, 14, 21 and 42) were evaluated with Friedman test, and to evaluate 2-time points, the Dunn test was performed. *Significant difference with baseline scores ($p<0.05$). δ Significant difference with difference with day 42 scores ($p<0.05$) (Friedman test, and to evaluate 2-time points, the Dunn test).

Figure legends

Figure 1. Flow diagram of the study design. PD: Probing depth; GR: Gingival Recession; mGI: modified Gingival Index; ABS: Angulated Bleeding Score; PI: Plaque Index; GCF: Gingival Crevicular Fluid; PICF: Peri-Implant Crevicular Fluid.

Figure 2. Changes in the inflammatory (A-G) and bone (H-K) cytokines in the crevicular fluid in teeth and implants of H-Health and H-GAP groups. Box plots represent the interquartile range, with the middle line representing the median and the whiskers showing the range. The upper brackets show P values from statistical tests that evaluated differences during the experimental protocol. The box below each graph shows comparisons between groups (H-GAP versus H-Health), using the Mann-Whitney test. Differences between tooth and implant in the same group was performed using the Wilcoxon test, and the differences are highlighted in the graphs (*= $p<0.05$; **= $p<0.01$; ***= $p<0.001$). δ indicates a significant difference in values compared to day 0 and ϵ indicates a significant difference in values compared to day 42 (Longitudinal analysis - Friedman and Dunn's tests). Significance level 95% ($p<0.05$).

Figure 3. **A** - α -diversity (Shannon index) of tooth and implant, of both groups before (Day 0) during (days, 7, 14, 21) and after biofilm accumulation. **B** - β -diversity (CTF metric) represented in the PCoA distance of tooth and implant, of both groups across time. **C** - β -diversity volatility (first axis - PC1) across time. **D** - Linear discriminant analysis (LDA) of β -diversity indices (a-d, Bray-Curtis; e-h, Jaccard; i-l, Unifrac Unweighted; m-p, Unifrac Weighted) over time in H-Health and H-GAP implants and teeth. No significant differences between groups (H-GAP versus H-Health) were observed. *Represents significantly difference between tooth and implants of the same group in that time point. If present, the differences among periods (longitudinal analysis) of each site are indicated by different letters (REML test, $p<0.05$).

Figure 4. LDA of the cytokine's levels and relative abundance of the top 10% differentials, in

tooth and implant, and during the healthy and inflammatory phase.

Appendices

Supplementary material

Table S1. Demographic and clinical characteristics of the study population.

	H-Health (n=10)		H-GAP (n=10)	p
<i>Age (Years $\pm$ SD)</i>	45.0 $\pm$ 10.1		41.7 $\pm$ 4.8	0.3834
<i>Gender (%F)</i>	70%		90%	0,5820
<i>Implant position – (%inferior arch)</i>	70%		50%	0,6499
<i>Implant area - (%anterior arch)</i>	0		40%	0,0867
<i>Time (month) in function (mean $\pm$ SD)</i>	63,9 $\pm$ 18,2		61,8 $\pm$ 23,4	0,9252
Clinical Parameters	Implant		Implant	Tooth
<i>Probing depth (mm)</i>	2.02 $\pm$ 0.97	1.86 $\pm$ 0.40	2.23 $\pm$ 0.67	2.03 $\pm$ 0.51
<i>Gingival/Mucosal Recession (mm)</i>	0 $\pm$ 0	1 $\pm$ 0.94	0 $\pm$ 0	1.72 $\pm$ 0.91

For the comparison between H-Health and H-GAP groups, a t-test was used for the parametric data (Age), and the Mann-Whitney test for the non-parametric data (Time in function) was applied. Fischer's exact test was used for non-numerical data. Significance level 95% ($p < 0.05$).